MASTER GUIDE: SOFTWARE ENGINEERING ROLES

Full Responsibility Taxonomy & Financial Governance Matrix for Individual Capstones INDUSTRY SDLC STANDARD

In industry software engineering, complex applications are built by specialized cross-functional roles. When students undertake individual capstone projects, they operate as a **One-Person Software Engineering Team**, systematically rotating through all 12 core development, operational, and managerial roles across the project lifecycle.

1. MASTER TAXONOMY OF ALL 12 SOFTWARE ENGINEERING ROLES

Domain	Role Title	Primary Industry Focus	Key Capstone Deliverable
Product & Leadership	**Product Manager (PM/PO)**	Product vision, monetization engines, feature prioritization.	Business & Revenue Model Pitch
	Project Manager (PM)	Schedule tracking, cost budgets, labor hours, risk control.	SRS Agreement & Dated Diary Log
	Business Analyst (BA)	Requirement gathering, client user stories, scope limits.	User Stories & Acceptance Matrix
Design & UX	**UI/UX Designer**	Visual component design, wireframing, design token systems.	Figma Prototype & Design System
	Information Architect (IA)	Content layout hierarchy, navigation trees, user journey flow.	Sitemap & Navigation Flowcharts
Engineering	**Front-End Developer**	Semantic HTML5 markup, responsive CSS3 grids, JS interactivity.	Git Source Code Repository
	API / Integration Engineer	Asynchronous `fetch()` data calls, JSON state, `localStorage`.	REST Endpoint Data Logic
	AI / Innovation Engineer	Client-side ML integration, WebAssembly AI execution.	Offline Transformers.js Chatbot
QA & Standards	**QA / Test Engineer**	Test suite execution, edge case form validation, bug logs.	Test Suite Matrix & Edge Case Logs
	Accessibility Specialist	WCAG 2.1 compliance, keyboard focus states, screen readers.	Lighthouse a11y Audit Scorecard
Operations & Security	**DevOps / Release Engineer**	Git branching workflows, automated CI/CD pipelines, DNS/SSL.	Production HTTPS Live Web URL
	Security Analyst (AppSec)	Input sanitization (XSS defense), `.env` security, HTTPS audit.	Security Vulnerability Report

2. FINANCIAL MANAGEMENT ALLOCATION: COST VS. REVENUE GOVERNANCE

Project Manager — Cost Management (Outflow)

- **Labor Cost Tracking:** Estimates and logs development effort based on hourly rates (e.g., 72 total hours × $15/hr).
- **Scope Creep Prevention:** Identifies new client feature requests and bills them as additional effort.
- **Infrastructure Budgeting:** Monitors recurring operational expenses (Domain, hosting bandwidth caps, API quotas).

Product Manager / Owner — Revenue Management (Inflow)

- **Monetization Architecture:** Selects the business model (B2B SaaS, Freemium, Transaction Fees, Affiliate links).
- **Pricing Strategy:** Defines pricing tiers (Free vs. $19/mo Pro Tier) and unit economics.
- **Monetization UI Design:** Implements client-facing pricing tables, upgrade modals, and order checkout summaries.

3. CAPSTONE LIFECYCLE PHASE & ROLE ROTATION MATRIX

Phase 1: Requirements Elicitation & Scope ——> *Roles: Business Analyst + Project Manager*
Phase 2: Product Architecture & Monetization ——> *Roles: Product Manager + Information Architect*
Phase 3: Visual Design & UI Prototyping ——> *Roles: UI/UX Designer*
Phase 4: Core Implementation & API Integration ——> *Roles: Front-End Developer + Integration Engineer*
Phase 5: Advanced Intelligence Features ——> *Roles: AI / Innovation Engineer*
Phase 6: Quality Assurance & Accessibility Audit ——> *Roles: QA / Test Engineer + Accessibility Specialist*
Phase 7: Cloud Deployment & Security Verification ——> *Roles: DevOps Engineer + Security Analyst*
Phase 8: Client Hand-off & Final Presentation ——> *Roles: Project Manager + Product Owner*